DIPLOMA PROGRAMME PRACTICE · SL · SET A

Biology / Paper 1A

Question booklet · 30 marks · 90 minutes across 1A and 1B

Answer every question. Choose one option, A–D, per question. Use a calculator. Record choices online or on separate paper.

QUESTION 1 [1]

An insect lifts its foot from a pond surface. A detergent reduces the upward force at the same contact length. Which property of water best explains the original force?

A Water adhesion to the foot alone, independently of attraction between water molecules
B The density difference between liquid water and the air above it
C Cohesion among water molecules at the interface
D High specific heat capacity, which resists local warming by the insect

QUESTION 2 [1]

A double-stranded DNA fragment contains 180 base pairs, including 54 G–C pairs. How many thymine nucleotides does it contain?

A 126
B 72
C 54
D 252

QUESTION 3 [1]

A cell is 18 µm wide. Its image measures 7.2 mm on the screen. What is the linear magnification?

A ×0.4
B ×40
C ×2500
D ×400

QUESTION 4 [1]

Two isolated bird populations look alike. Which observation most directly supports treating them as separate biological species?

A They occupy forests on different islands
B When brought into contact they repeatedly fail to produce fertile offspring despite normal breeding conditions
C Their mean adult masses differ by 3%
D They have different frequencies of a plumage allele

QUESTION 5 [1]

A fossil mammal has a forelimb with the same sequence of bones as a bat wing, but different bone proportions. Which inference is best supported?

A The limbs may be homologous despite different functions
B The fossil must be a direct ancestor of bats
C The limbs must have evolved independently because their shapes differ
D The two species necessarily occupied identical niches

QUESTION 6 [1]

An island breeding programme can house only 20 founders. Which selection best protects genetic diversity across the source population?

A Choose individuals expressing one preferred colour
B Use ten sibling pairs that already breed successfully
C Choose the twenty heaviest adults from one family
D Sample minimally related individuals across the source range

QUESTION 7 [1]

Two polymers yield glucose on complete hydrolysis. Human amylase acts on one but not the other. Which difference best explains this?

A The arrangement of glucose is identical, but amylase recognizes the free chain ends only
B The orientation of the glycosidic linkages differs
C The number of glucose residues alone determines whether either polymer can bind amylase
D Cellulose is resistant because all bonds connecting its chains are covalent

QUESTION 8 [1]

After heating, a protein loses activity. Chemical analysis detects no change in amino acid sequence. Which conclusion is justified?

A A changed side-chain sequence is the only way to alter active-site shape
B Loss of activity demonstrates cleavage of the polypeptide backbone into separate chains
C The active site must remain functional as long as peptide bonds remain intact
D The functional three-dimensional conformation may have changed

QUESTION 9 [1]

Cells equilibrate with a dye through a specific channel. ATP production is inhibited but a dye concentration gradient remains. What should occur while the channel stays open?

A Net movement against the gradient using the channel alone
B Movement stops because all membrane transport requires ATP
C Dye moves only after vesicles fuse with the membrane
D Net movement down the dye concentration gradient

QUESTION 10 [1]

A gland cell secretes a polypeptide enzyme. Which route would a newly made enzyme normally follow?

A Nucleus → lysosome → chloroplast
B Rough ER → Golgi apparatus → secretory vesicle
C Smooth ER → mitochondrion → nucleus
D Golgi apparatus → nucleus → rough ER

QUESTION 11 [1]

Two similarly shaped cells have widths of 10 and 20 μm. The larger cell has how much membrane area per unit volume compared with the smaller cell?

A Twice as much
B The same amount
C One half as much
D Four times as much

QUESTION 12 [1]

Fibrosis doubles the diffusion distance across a gas exchange surface. All other factors remain fixed. How does the initial oxygen diffusion rate change under an inverse-distance model?

A It is unchanged
B It falls to one quarter
C It halves
D It doubles

QUESTION 13 [1]

A valve between the left ventricle and aorta opens. Which pressure relationship and direction of flow apply at that instant?

A Atrial pressure exceeds venous pressure and blood enters the aorta directly
B Equal pressure across the valve actively pumps blood forward
C Ventricular pressure exceeds aortic pressure and blood moves into the aorta
D Aortic pressure exceeds ventricular pressure and blood moves into the ventricle

QUESTION 14 [1]

Two desert plants have equal leaf area. Plant X has stomata recessed into still-air pockets. Which consequence is most likely in dry wind?

A X can take up carbon dioxide only at night
B A more humid local boundary layer reduces the water-vapour gradient
C X no longer requires xylem
D Water leaves faster because the diffusion path shortens

QUESTION 15 [1]

A grazer usually occupies an entire shore when its competitor is excluded. With the competitor present it occupies only the lower shore. What do the results suggest?

A Both species occupy identical realized niches everywhere
B Competition expands the fundamental niche
C The upper shore is always physiologically intolerable
D Competition restricts the realized niche

QUESTION 16 [1]

A substrate disappears from 8.0 to 6.2 mmol in 30 s. A matched tube lacking enzyme loses 0.30 mmol. What is the background-corrected mean enzymatic rate?

A 0.070 mmol s^{-1}
B 0.050 mmol s^{-1}
C 0.010 mmol s^{-1}
D 0.060 mmol s^{-1}

QUESTION 17 [1]

Germinating seeds warm an insulated flask while an equal mass of boiled seeds does not. Which explanation best connects the observation to respiration?

A All energy from substrates is stored indefinitely as ATP
B Heat is produced only by photosynthesis
C Some chemical energy released from substrates is transferred as heat
D Oxygen supplies the carbon atoms of respiratory substrates

QUESTION 18 [1]

An illuminated algal culture releases oxygen at a net rate of 8 units per hour and consumes oxygen in darkness at 3 units per hour. Assuming the same respiration rate in light, what is its gross photosynthetic oxygen production?

A About 2.7 units per hour
B 5 units per hour
C 8 units per hour
D 11 units per hour

QUESTION 19 [1]

A sensory neuron carries an impulse from a receptor toward the spinal cord. Which change moves along its axon?

A Movement of the receptor cell to the spinal cord
B A wave of membrane depolarization
C Bulk flow of neurotransmitter inside the axon
D Diffusion of an action potential through blood

QUESTION 20 [1]

A bright light reaches one eye and its pupil becomes smaller. Which sequence best represents the reflex pathway?

A Receptor → sensory neuron → central integration → motor neuron → effector
B Receptor → motor neuron → sensory neuron → effector
C Central integration → receptor → sensory neuron → effector
D Effector → motor neuron → receptor → sensory neuron

QUESTION 21 [1]

After a booster dose, antibody concentration reaches a higher peak sooner. Which mechanism best explains this?

A Antigen-specific plasma cells arise from erythrocytes during the booster response
B Primary antibodies cause all later antigens to be removed before recognition
C Memory lymphocytes initiate a rapid specific response
D Memory B cells produce a faster response by lowering the specificity of antigen recognition

QUESTION 22 [1]

A closed population of beetles has 80 marked adults. A later unbiased sample contains 12 marked and 48 unmarked adults. What population estimate follows from equal capture probabilities and thorough mixing?

A 320
B 400
C 100
D 480

QUESTION 23 [1]

Primary consumers ingest 600 kJ from plants and add 54 kJ to their own biomass. What percentage of ingested energy becomes new consumer biomass?

A 90%
B 54%
C 11.1%
D 9%

QUESTION 24 [1]

Both parental DNA strands carry a heavy label. Replication occurs twice in light medium. What fraction of double-stranded molecules still contains a heavy strand?

A One half
B One quarter
C None of them
D All of them

QUESTION 25 [1]

A coding DNA strand reads 5′-ATG GAA TTT-3′. Which mRNA sequence is transcribed from the complementary template?

A 5′-AUG GAA UUU-3′
B 5′-ATG GAA TTT-3′
C 5′-AAA UUC CAU-3′
D 5′-UAC CUU AAA-3′

QUESTION 26 [1]

A substitution changes a coding triplet but leaves the amino acid unchanged. Which statement explains the result?

A The ribosome reads overlapping triplets and thereby corrects a changed base
B Different codons can specify the same amino acid
C A substitution changes mRNA but can never affect the encoded amino acid
D All tRNAs carry more than one amino acid at a time

QUESTION 27 [1]

A diploid cell contains 12 chromosomes in G1. Immediately after DNA replication and before mitosis, how many chromosomes and chromatids are present?

A 24 chromosomes and 24 chromatids
B 6 chromosomes and 12 chromatids
C 12 chromosomes and 24 chromatids
D 12 chromosomes and 12 chromatids

QUESTION 28 [1]

Plant tissue gains 6% mass in solution X and loses 3% in solution Y. Samples began in the same condition. Which inference is supported?

A X favours net water entry and Y favours net water loss
B The tissue water concentration was identical to both solutions
C Both solutions are isotonic with the tissue
D Solute must have moved by active transport in both

QUESTION 29 [1]

A heterozygote with a dominant phenotype is crossed with a homozygous recessive individual. What is the probability that two successive offspring both show the recessive phenotype?

A ¼
B ½
C ¾
D ⅛

QUESTION 30 [1]

Dark beetles survive better on dark ground. After several generations their frequency increases. What further observation most directly connects this to natural selection?

A Colour is heritable and dark survivors leave more surviving offspring

B Every beetle has an equal number of offspring

C Predators always prefer the most abundant beetle

D Individual beetles darken whenever predators appear